

Longitudinal Analysis of Forward Blind Zone Changes in Popular Vehicle Models (1997–2023)

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Abstract

Background. In 2022, vulnerable road user (VRU) deaths in the United States increased to their highest level in more than 40 years. At the same time, increasing vehicle size and taller front ends may contribute to larger forward blind zones, but little is known about the role that visual occlusion may play in this trend. **Goal.** Researchers measured the blind zones of six top-selling light-duty vehicle models (one pickup truck, three SUVs, and two passenger cars) across multiple redesign cycles (1997–2023) to determine whether the blind zones were getting larger. **Method.** To quantify the blind zones, the markerless method developed by the Insurance Institute for Highway Safety was used to calculate the occluded and visible areas at ground level in the forward 180° arc around the driver at ranges of 10 m and 20 m. **Results.** In the 10-m forward radius nearest the vehicle, outward visibility declined in all six vehicle models measured across time. The SUV models showed up to a 58% reduction in visibility within a 10 m radius. Other vehicles exhibited smaller (7%–19%) reductions. At longer distances (10 m–20 m), vehicles demonstrated both increases and decreases in visibility. **Conclusion.** The markerless method provides a straightforward and replicable assessment of driver visibility. The observed decrease in direct outward visibility near the vehicles points to the need for further study regarding this trend, including analysis of the repeatability and viability of the measurement technique.

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1. Introduction

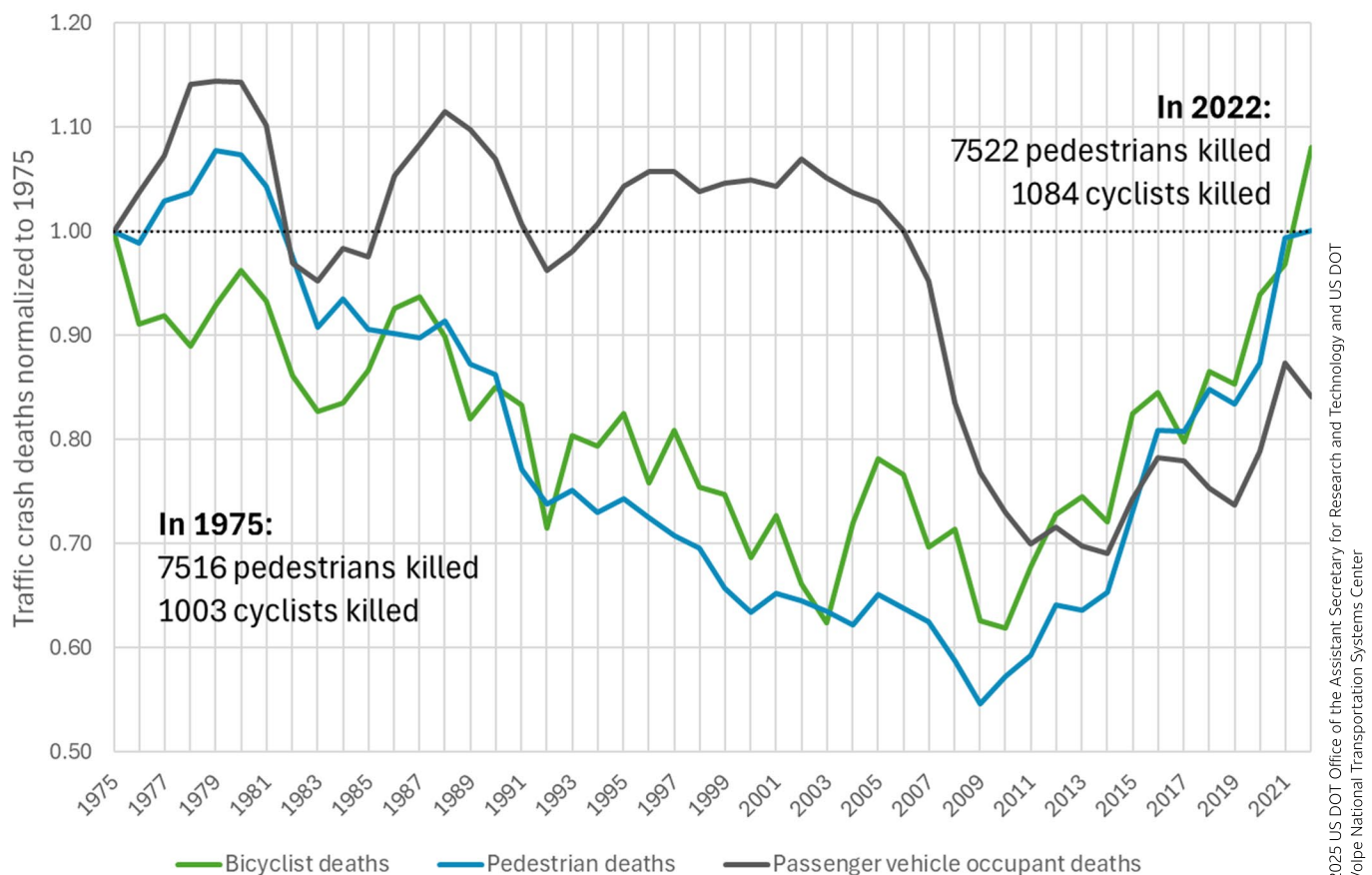
1.1. Background

There has been a widely acknowledged and documented increase in vulnerable road user (VRU) fatalities in the United States. Pedestrians killed in crashes involving passenger vehicles have rapidly increased over the model years measured in this study, between 1997 and 2022, increasing 41% from 5306 in 1997 to 7522 in 2022. The number of people biking who were killed in vehicle crashes has shown an equally dramatic change, increasing 36% from 1997 to 2022 [1]. Of the 7522 pedestrian fatalities in 2022, 3388 involved a light truck [1], 814 of these fatalities were at an intersection or were intersection-related. In large trucks, it has been argued that visual obstruction contributes to pedestrian strikes in low-speed maneuvers, such as at intersections [2].

While pedestrian and bicyclist deaths have risen rapidly in the study period (1997–2022), passenger vehicle occupant fatalities have declined during the same study period, providing a motivation for the study period bracketing the inversion on both sides. Taking the long view, note that both pedestrian and bicyclist deaths in 2022 exceeded those in 1975, the first year of federal record-keeping, while vehicle occupant deaths declined 16% over the same period—see [Figure 1](#). VRU deaths also reached a record high of over 20% of all traffic crash deaths in 2022 [3, 4].

A number of factors could explain the rise in VRU fatalities, including changes in vehicle design, driver behavior, and road infrastructure. Because extensive research has explored behavioral issues such as distraction and impairment as causal factors in pedestrian and bicyclist fatalities [5, 6], we do not consider these aspects in this work and instead focus on factors related to vehicle design ([Section 1.2](#)). Much the same holds true for the relation between road infrastructure and pedestrian and bicyclist fatalities [7].

FIGURE 1 Pedestrian, bicyclist, and passenger vehicle occupant deaths in the United States, 1975–2022. (Data taken from Ref. [3, 4].)



1.2. Vehicle Design

When considering surface transportation safety, there are two different domains of development that work in tandem: collision avoidance and crash mitigation. Before a crash occurs, the core objective is to prevent an impact through driver action or automatic intervention (e.g., automatic braking). After a collision is certain to occur, the emphasis shifts to crashworthiness and mitigation, where the objective is to minimize harm to the individuals involved in the incident. In this layered approach the first objective is to avoid a crash, while mitigation factors are in place as a backstop. Consideration of both these domains plays a role in the design of a vehicle.

The observed rise in VRU fatalities occurs during a time when light trucks/vans (pickups, SUVs, and minivans) made up an increasing proportion of the vehicle on-road fleet, but light trucks/vans are also the striking vehicles in a disproportionately high fraction of VRU fatalities [8]. Specifically, one Illinois study found that SUVs struck pedestrians and cyclists in 14.7% of all crashes but were involved in 25.4% of the fatalities [8]. This same study found that pickup trucks hit pedestrians and cyclists in 5.6% of the crashes but were involved in 25.4% of the fatalities. Another study found that the odds of pedestrian fatalities when the pedestrian was struck by light trucks was 1.54 times the odds of pedestrian fatalities when the pedestrian was struck by passenger cars [9]. This disparity suggests that vehicle attributes are an important factor in harm to VRUs.

There are a variety of reasons why light trucks might be more lethal to pedestrians and bicyclists than passenger cars, and the design of both passenger cars and light trucks/vans has evolved substantially. An Insurance Institute of Highway Safety (IIHS) analysis [10] found that the average US passenger vehicle has become about “4 in. wider, 10 in. longer, 8 in. taller and 1000 lb heavier over the last 30 years.” Consumer Reports found that the hood height of light trucks/vans has increased by an average of at least 11% since 2000 and that new pickups grew 24% heavier on average from 2000 to 2018 [11]. These changes impact both crash mitigation and crash avoidance.

Within the crash mitigation domain, Hu et al. [12] used a combination of medical records and crash records to demonstrate that taller leading edge (hood) heights and more vertical profiles are linked to more frequent head strikes and more severe pedestrian injury outcomes. More recently, 202 pedestrian crashes were analyzed to estimate the moderating effect of the hood leading edge height on the relation between impact speed and the severity of injury. The hood leading edge height significantly increased the risk of pedestrian injury at all speeds [13]. Within the crash avoidance domain, taller hoods also decrease forward visibility, intrinsically influencing a driver’s ability to detect people or objects in their vehicle’s path. Seat position, the size and shape of the A-pillars

and B-pillars [14, 15], and the increase in the leading edge and hood height of vehicles [12] can all change the size and position of areas that are visible and non-visible to drivers.

In this work, we focus on the crash avoidance domain by exploring the role of vehicle shape in a driver’s opportunity to see people or objects around them. Specifically, no known research has conducted repeated measurements of the visually obstructed area in a 180° forward arc around the driver (for short, the blind zone) of specific makes and models (nameplates) of light trucks/vans and passenger cars over time. Such an analysis could be used to establish the exact relation between the size and location of vehicle blind zones and crashes and provide information for possible countermeasures.

In the remainder of the Introduction, we begin by reviewing studies relating the blind zone to actual and simulated crashes. We then describe the various methods that have been used to measure blind zones.

1.3. Blind Zones as a Factor in VRU Crashes

Drivers’ visibility to the front, sides, and rear directly determines how well they can see vulnerable road users and other vehicles. Although drivers’ visibility through the rear of their vehicle and its corresponding crash risk has been addressed with surface vehicles of all types [16, 17], the relation between crash risk and drivers’ direct visibility of the roadway to the front and sides has been studied primarily with commercial motor vehicle (CMV) drivers [18]. This relationship has been studied most extensively in the United States and Europe.

In the United States, both field and simulator studies have been used to document the risks presented by large blind zones of CMVs. Using field data, it has been estimated, in the United States, that 20% of the crashes involving heavy trucks are caused by visibility limitations [19]. Specifically, most low-speed maneuver (LSM) crashes of heavy vehicles involve obstructions to drivers’ vision (geometric obstruction) [2]. More recently, a simulator study showed that drivers in low-vision truck cabs (mirrors were necessary to see to the side and front) were much more likely than drivers in high-vision truck cabs (direct vision could be used to see to the side and front) to strike pedestrians crossing in front of them at an intersection [20]. The estimated blind zone volume of the simulated low-vision cab within 15 ft to the front and right side and within 6 ft above ground level was 67% (2017 Mack Granite); the blind zone volume of the simulated high-vision cab was 22% (2020 Dennis Eagle). In the intersection scenario, the driver was stopped at a four-way signalized intersection. Simulated pedestrians crossed from the left and the right and were visible to the driver through the side windows. One of the crossing pedestrians stopped in front of the truck, just as the signal turned from red

to green. In 87% of the simulated scenarios, the driver in the low-vision truck cab struck the pedestrian. In the same scenarios, none of the drivers in the high-vision truck cab struck the pedestrian. In the most easy-to-track scenario, only one pedestrian crossed from the left to the right. Importantly, eye glance behaviors indicated that the driver looked at the crossing pedestrian. Yet 67% of the drivers in the low-vision truck still accelerated forward when the signal turned green and struck the pedestrian that had stopped in front of them.

Much the same conclusions have been reached in studies conducted in Europe. Using field data, it was argued that the lack of direct vision to the front and sides of heavy goods vehicles (HGVs) leads to VRU strikes [21]. The strikes were particularly problematic for HGVs because the box-like shape of the HGVs tended to knock the VRU onto the ground and under the wheels of the HGV. Additionally, research on a truck simulator indicated that drivers in low-vision truck cabs were much more likely to strike VRUs than drivers in high-vision designs, where daylight openings are larger, these openings extend lower, and the driver seating position is lower [22]. Drivers in the low-vision truck cab, which has a high seating position and smaller, higher daylight openings, could see the pedestrian crossing in front of the truck with Class VI mirrors, similar to crossover mirrors in the United States. Yet, the study found 26.7% of participants in the low-vision truck collided with a VRU, versus 3.3% in the high-vision truck.

In summary, studies in the field and using simulators, both in the United States and Europe, indicate that the size of the blind zone of CMVs and HGVs as determined by the cab design are significant contributors to forward and turning crashes. However, no prior studies have systematically measured changes in blind zones across multiple vehicle generations. The next question is how to measure the size of the blind zone.

1.4. Measurement of Blind Zones

Various methods have been used to measure visual obstruction to the front and sides of vehicles (the blind zone). Some methods that have been researched over the years have advanced proofs of concept but have not been widely deployed for decision support due to either accuracy or standardization challenges or lack of maturity scaling of the method. Two examples include photogrammetry of a panoramic field-of-view photograph from the driver's eyepoint [23] and the use of consumer LiDAR to create a digital representation of the vehicle [24, 25]. Three measurement methods have been more widely deployed to date for decision support: the TfL and UNECE method [26–29], the linear cone method [30], and the IIHS Areal method [31]. These methods have benefits and drawbacks depending on the level of accuracy and effort required to collect the data.

1.4.1. Volumetric: TfL and UNECE Perhaps the most standardized and widely adopted method is the volumetric assessment approach, which has now been formalized into regulation in Europe. Transport for London (TfL) [26] together with the Transport Research Laboratory [27] and the University of Loughborough [28] developed this method in 2016, which analyzes CAD models of heavy-duty trucks to precisely quantify blind zones within a defined assessment volume proximate to the truck cab [26, 32]. The volumetric approach of the Direct Vision Standard was expanded, using a physical measurement approach based on a standardized eyepoint rig and a ground-plane measurement grid, into the United Nations Regulation No. 167 for Commercial Vision Direct Vision, adopted November 2022 [29]. The assessment volume is based on the same ground area covered by indirect vision as regulated for heavy vehicles by UN Regulation No. 46, but extended with 2 m on the driver's side. The height of the assessment volume is determined by the shoulder height of the tallest European population. Challenges in applying this approach at scale in the United States include the difficulty of acquiring accurate CAD models across a wide selection of vehicles, as this would depend either on cooperation from vehicle manufacturers or a relatively labor-intensive data collection process for the physical method.

1.4.2. Linear: Cone Method In the United States, where the direct vision of vehicle design is not federally regulated, the City of Boston and Together for Safer Roads, a global NGO [33], have implemented a simplified, low-resource approach, the “cone method.” In contrast to the three-dimensional volumetric approach, this one-dimensional method is simplistic, but it is straightforward to validate. It measures the minimum distance at which a median height male driver can see a 36-in. or 48-in.-tall traffic cone or other object, corresponding, respectively, to the approximate shoulder height of 5th-percentile female US elementary school children and adults. The cone method collects the distance to the cones in the forward and passenger-side directions [34]. A tradeoff of this approach is that it only quantifies the blind zone in those two cardinal directions, not in other directions where pedestrians may be occluded from the motorist's view before a crash, in exchange for being broadly accessible and replicable without advanced equipment or training.

1.4.3. Areal: IIHS Method The method that this article focuses on is the “markerless method.” This two-dimensional approach, developed by the IIHS and modified in collaboration with USDOT Volpe Center, has recently been deployed to inform MassDOT's direct vision planning efforts [30]. In brief, the IIHS method offers a compromise between the previous methods. The IIHS camera-based technique assesses the blind zone area at ground level (or, in the next version in development, at pedestrian shoulder heights or other planes of interest) in an arc of up to 360° around the driver and at any distance of

interest from the driver eyepoint position [35]. The resulting data is comparable to data produced by the labor-intensive shadow tracing method of ISO 5006 [36]. It is possible for anyone to replicate the markerless process so long as they have a camera, a standardized rig, and access to the vehicle. A validation by IIHS found all points on the ground to the front and sides of the vehicle to be within 0.5 m of ground truth measurements [31].

2. Method

2.1. Vehicle Selection

The research team identified top-selling pickup trucks, SUVs, and sedans from 1990 to present, along with the years where major redesigns of each model occurred. Using these schedules, researchers sought to measure representative vehicles from each major redesign. The team was able to measure 17 vehicles from 6 top-selling light vehicle nameplates (Toyota Camry, Honda Accord, Honda CR-V, Chevrolet Suburban, Jeep Grand Cherokee, and Ford F-150) with model years from 1997 to 2023.

2.2. Field Measurement

The research team followed a standardized field measurement protocol based on a modified version of the IIHS Driver Direct Visibility Protocol Version 1 [31]. The protocol is summarized as follows (Figure 2):

1. Set the driver seat into a precise mid-height, mid-track position, relying on a Bosch laser range finder and target for positioning.
2. Place a camera rig atop the driver's seat that positions the lens at a standardized cyclopean eyepoint for a 50th-percentile male driver. (While the rig included a swappable anthropometric neck that can replicate a 5th-percentile female driver, a 50th-percentile male driver, and 95th-percentile male driver, for the purposes of this study, all measurements were completed with the 50th-percentile male driver eye point.)
3. Collect photographs, sweeping left to right at 30° increments, to cover the field of view in the forward 180°, noting the camera position measurements.

2.3. Visibility Measurement: IIHS Markerless Method

Given the advantages of the markerless method for the purposes of this research, the report uses the first version of an open-source software developed by IIHS in 2023 and licensed under the terms of the Apache License,

FIGURE 2 Rig used for markerless method.



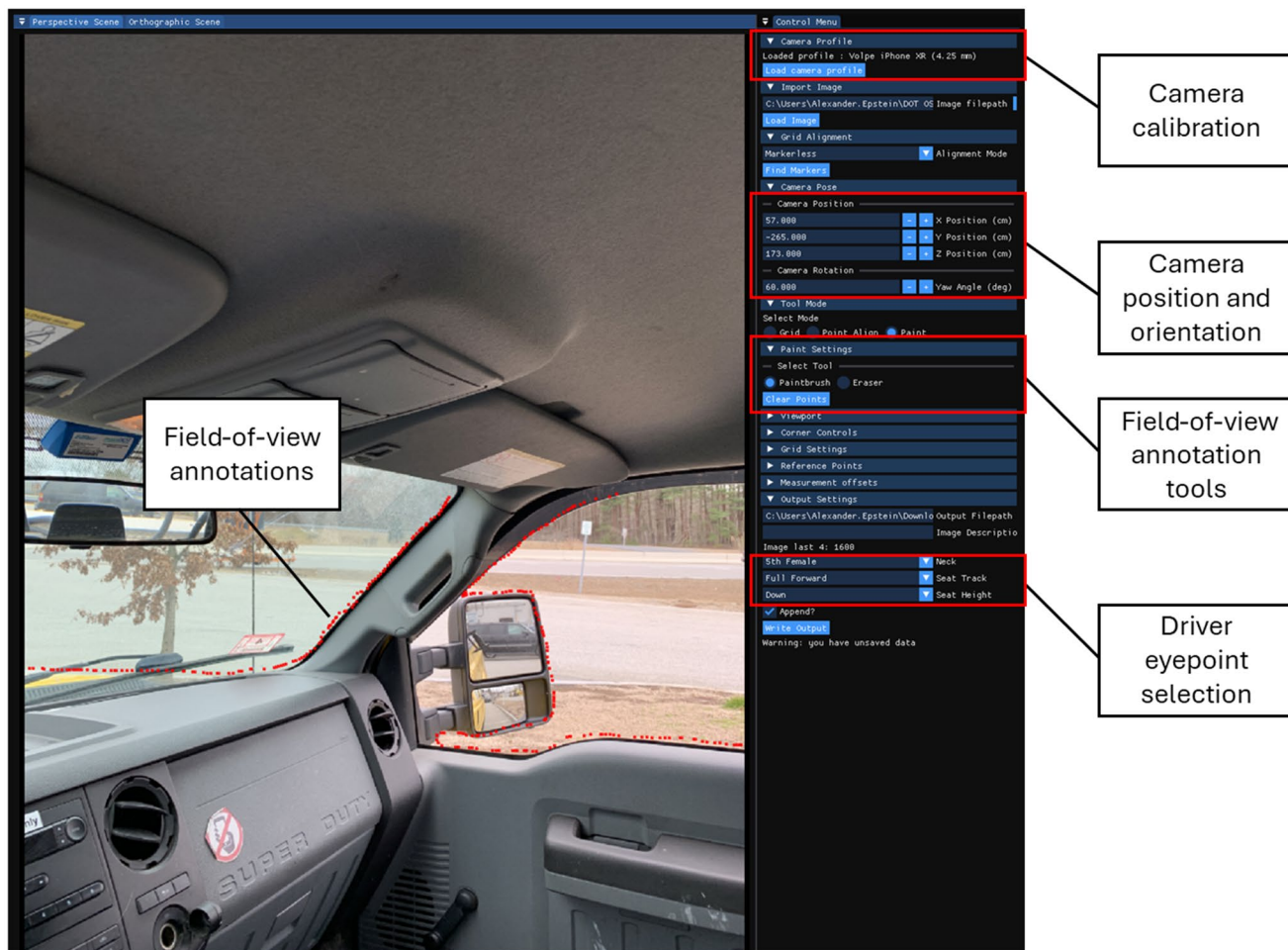
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version 2.0 [31]. The software prototype is a C++ application (app) that allows users to input images and measurements of the camera in reference to the front corner of the vehicle to project the nearest visible ground points onto a virtual cartesian coordinate system. In actual practice, photographs are imported into the app and manually annotated by the user. The user draws the field of view along the lower edges of the vehicle windows, mirrors, and A-pillars (Figure 3). Results from each photo are exported into a CSV file for merging and further analysis.

Specifically, this application uses a perspective transformation method to compute the blind zone area at ground level from the above data. In addition, the method requires knowledge of camera properties. These properties are obtained from a series of camera calibration frames and x , y , z , θ coordinates of the camera relative to the origin of the left front corner of the vehicle at ground. This information is then used to apply a virtual grid around the vehicle [37]. No external markers or grids are required in the photograph for calculation.

2.4. Data Processing

The outputs from the version of the IIHS markerless software used from this study required additional processing to create the final results for analysis. (More recently, the software has been updated to improve the post-processing methods [31].) First, the research team mapped the x , y coordinates of the nearest visible ground points in ArcGIS Pro, used under a commercial software license, and removed outlier data points that were clearly non-physical artifacts and not part of the field-of-view curve. As the input photographs overlap, the nearest

FIGURE 3 Screenshot of IIHS markerless perspective transformation software.

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visible points also overlap. The research team removed overlapping data points, generally keeping the nearest data points where overlaps occurred. Consistent with vehicle stopping distances in the low-speed maneuvers typical of blind zone crashes involving people walking or biking [2, 38], the results in this study defined assessment blind zone areas as 10 m, 15 m, and 20 m semicircles centered on the driver eyepoint. The research team removed any nearest visible points beyond these distances.

As a final step, the research team connected the coordinate points to create a polygon of blind zones around the vehicles. These extended to the 10 m, 15 m, or 20 m radii at areas of the vehicle that are completely obscured, such as the A-pillars.

3. Results

Table 1 shows the visible area results among the six top-selling, light-duty vehicles (comprising a total of 17 vehicles) measured between model years 1997 and 2023. Looking

within a 10 m boundary, all vehicles showed a decrease in visibility from earlier model years to the later model years measured. The Chevrolet Suburban and Honda CR-V showed the most dramatic decreases in visibility within a 10 m radius: the latest measured model years were found to have 49% and 42%, respectively, of the visible areas in their earliest measured model years. Based on the average driver stopping distance at 10 mph (27 ft or 8.2 m) [38], a low speed at which blind zones have been reported to be a common factor in crashes [2], the visible area within a 10 m radius was used as the primary comparison metric. However, the change in the visible area between a 10 m radius around the driver and a 20 m radius around the driver was also measured. It decreased slightly in some models and increased slightly in others (Table 1). Note that the 15 m results are available from the authors but are not reported here.

Plots of the visibility and the 10 m, 15 m, and 20 m semicircles are given in Figures 4–6. In all figures, the nearest visible points for different model years are represented by lines of different colors. We comment briefly on three of the visibility plots, showing that the visibility

TABLE 1 Visible area and percentage of area that is visible, by make, model, and year of the specific measured vehicle. Design years note the generation of the vehicle.

Make	Model	Model year	Design years	Visible area within 10 m radius (m ²)	Percent visible area within 10 m radius (m ²) (%)	Visible area between 10 and 20 m radius (m ²)	Percent visible area between 10 and 20 m radius (m ²) (%)
Chevrolet	Suburban	2006	2000–2006	88.5	56.4	416.6	88.4
Chevrolet	Suburban	2016	2015–2020	75.9	48.3	417.8	88.7
Chevrolet	Suburban	2023	2021–present	43.3	27.6	399.2	84.7
Ford	F-150	2001	1997–2003	68.2	43.4	410.6	87.1
Ford	F-150	2014	2009–2014	62.7	39.9	418.0	88.7
Ford	F-150	2015	2015–2020	56.3	35.9	410.4	87.1
Honda	CR-V	1997	1997–2002	106.3	67.7	420.4	89.2
Honda	CR-V	2009	2008–2012	90.7	57.7	404.2	85.8
Honda	CR-V	2014	2013–2017	69.3	44.1	410.6	87.1
Honda	CR-V	2022	2018–2024	44.2	28.1	385.3	81.8
Honda	Accord	2007	2003–2007	102.1	65.0	381.3	80.9
Honda	Accord	2021	2018–2023	93.6	59.6	411.0	87.2
Jeep	Grand Cherokee	2006	2005–2010	97.9	62.3	385.4	81.8
Jeep	Grand Cherokee	2020	2011–2021	70.8	45.0	395.4	83.9
Jeep	Grand Cherokee	2023	2022–2023	79.0	50.3	406.3	86.2
Toyota	Camry	2008	2007–2011	96.0	61.1	386.8	82.1
Toyota	Camry	2023	2018–2023	89.2	56.8	408.1	86.6

FIGURE 4 1997 and 2022 Honda CR-V overhead view showing nearest visible points at ground level for the forward 180°.

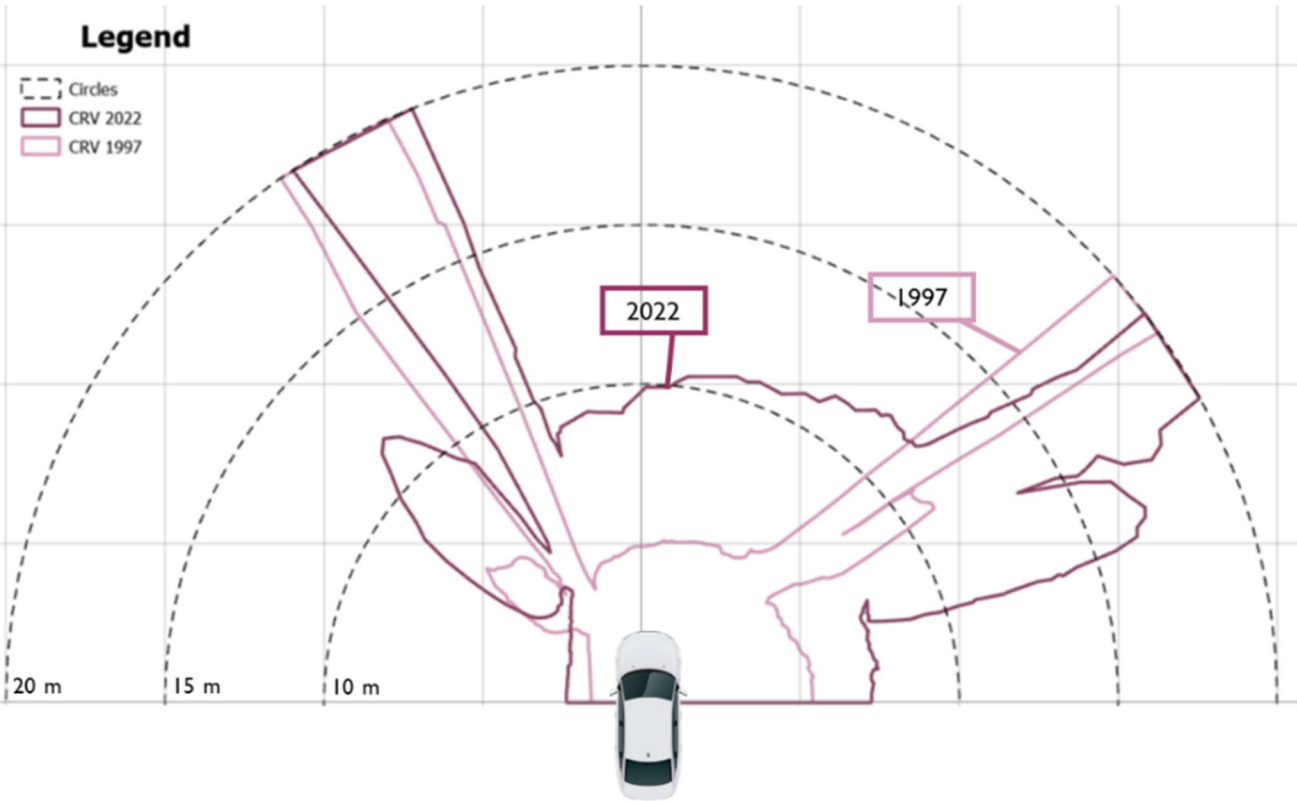
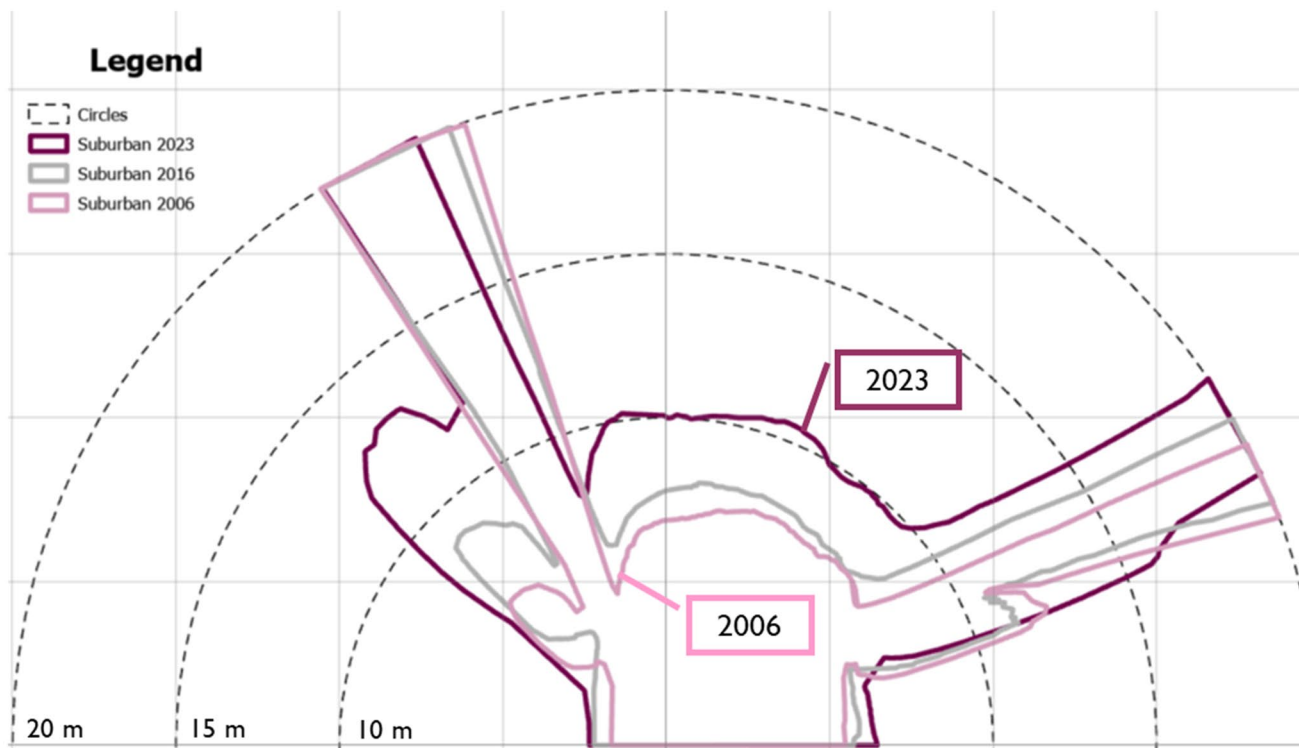
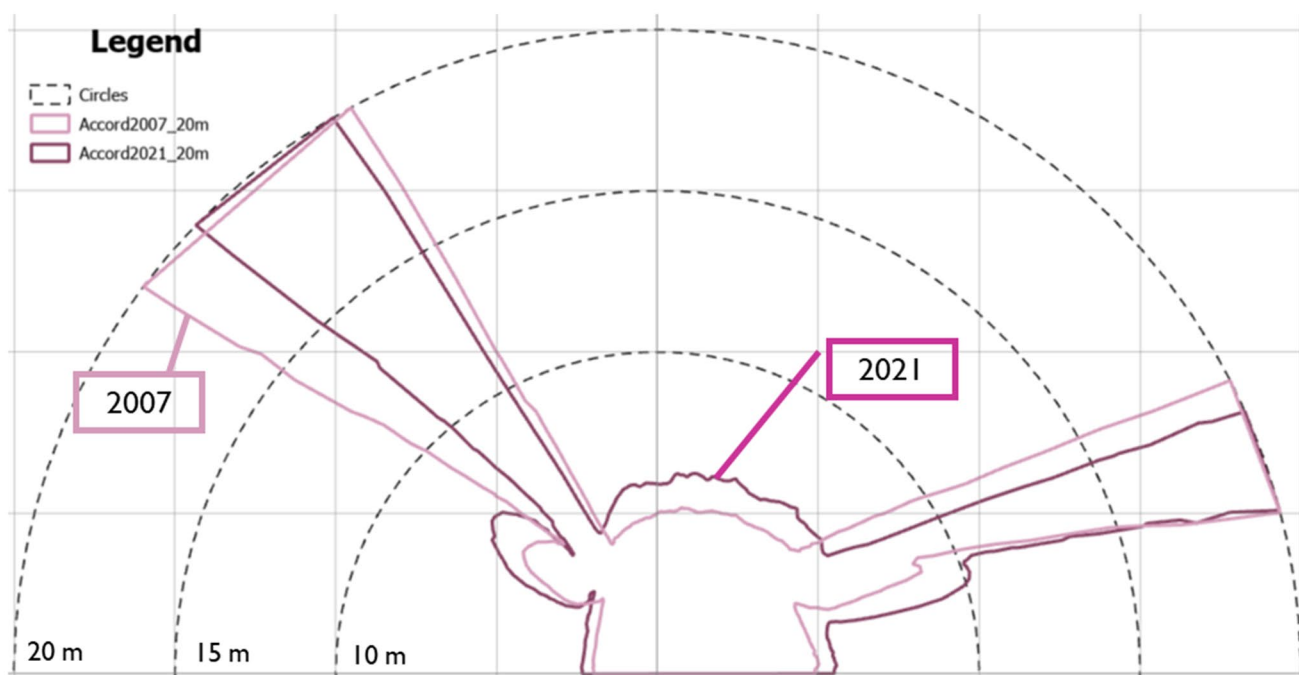


FIGURE 5 2006, 2016, and 2023 Chevrolet Suburban overhead view showing nearest visible points at ground level for the forward 180°.



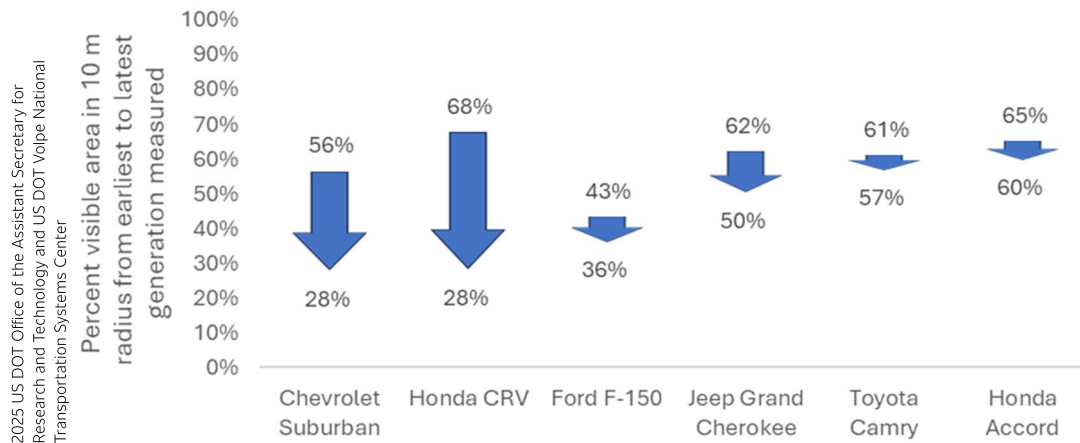
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FIGURE 6 2007 and 2021 Honda Accord overhead view showing nearest visible points at ground level for the forward 180°.



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FIGURE 7 Change in visible area within 10 m of the driver for vehicle models in the study, from the earliest to latest generations measured.



decreases in the 10 m semicircle blind zones can result from different changes in blind zone geometry. Consider first the Honda CR-V. In Figure 4, the Honda CR-V shows major increases in the ground-plane blind zone in the 2022 model year compared with the 1997 model year in most areas: across the front hood, larger mirrors, and passenger A-pillar. This was the largest observed change in visibility across the vehicles in this study. As another example, consider the Chevrolet Suburban. Like the Honda CR-V, the largest changes in the ground-level blind zone are across the frontal plane and the side mirrors (Figure 5). However, the impact of the driver-side mirror on the area of the blind zone is much more dramatic. In contrast, demonstrating one of the smallest changes in observed visibility was the Honda Accord. The visibility to ground across the hood is relatively similar when comparing the 2003–2007 generation against the 2018–2023 generation. The apparent minor decrease is partly created by larger side mirrors, as seen in Figure 6.

In summary, all six vehicles demonstrate varying degrees of decreasing visibility at ground level in the 10 m blind zone over the study period, as summarized in Figure 7.

- Honda CRV, Chevrolet Suburban, Jeep Grand Cherokee, and Ford F-150 demonstrate decreases, ranging from 16% to 58%.
- Toyota Camry and Honda Accord both exhibit minor decreases (7%–8%), which may fall within measurement error.

4. Discussion

The markerless measurement approach used in this work shows a feasible method that can be scaled to measure many vehicles, as long as each is physically accessible. Specifically, the markerless method does not require CAD

drawings of the vehicle, does not require that cones be physically laid on the ground, and does not require manual computation of the blind zone area. Moreover, the markerless method is repeatable and has been shown to be accurate.

This initial study of used vehicles in the field reveals that six of the top-selling vehicles in the United States all showed decreasing visibility in the 10 m forward ground-plane blind zone over the past 25 years. We believe that this is the first such study to show a change over time. The across-the-board decreases in near-field visibility observed in this study suggest that there could be a relation between visibility and fatality rates, given that (a) there is an intrinsic relationship between the ability to see something in one's path and react to it, (b) CMV simulator studies have demonstrated that decreased visibility is associated with increased pedestrian strikes [20], and (c) the shrinking visible areas around top-selling vehicles are directionally consistent with both the observed increase in roadway VRU fatalities and with increasing pedestrian fatalities in low-speed, non-public-road locations such as parking lots and driveways [39]. If this relation were to hold, it would point to the development of countermeasures that focused on increasing the driver's forward visibility, thereby potentially decreasing pedestrian fatalities.

However, to draw conclusions about a causal relationship between visibility and fatality rates, additional data and additional vehicle factors would have to be addressed, such as increased horsepower, sound insulation, and high seating position—which can encourage more frequent and severe speeding as well as decreased situational awareness [40, 41]. Increased hood height is also a factor [13]. Some of these factors may be covariates that correlate with one another and may interact synergistically. For example, larger blind zones associated with higher-seating positions, tall hoods, and wide pillars in light trucks/vans may reduce optic flow for drivers, inducing

higher driving speeds that increase crash risk while taller, more aggressive front-end designs increase fatality risk to VRUs [42].

Additionally, more information is needed about the relation between static measures of visibility and dynamic measures of visibility. When moving, objects that are not visible can become visible and objects that are visible can become invisible. A recent study on a driving simulator suggests that in right hook crashes which involve a vehicle turning right at an intersection and a cyclist traveling straight, the driver spends on average a total of only 350 ms glancing to the right out the passenger-side window for a bicyclist [43]. This is roughly the equivalent of a single glance. If this is the case, then the measure of static visibility may also capture the measure of dynamic visibility.

We have not discussed as of yet possible countermeasures that could increase visibility or studies that could better identify the exact role that visibility plays in crashes. In the remainder of this section, we discuss technological approaches that may mitigate the impacts of blind zones. We then conclude by suggesting future lines of research that may demonstrate any causal relationship that might exist.

4.1. Technological Countermeasures

While this study shows there is a measurable increase in the blind zones of top-selling light trucks/vans, it has been proposed that different technologies can compensate for, or at least mitigate, the safety impacts of increased blind zones. These include:

- Mirrors such as convex side mirrors and crossover or cross-view mirrors (primarily implemented on medium/heavy, not light trucks/vans) can help a driver see VRUs in the blind zone. However, mirrors have been reported to prolong drivers' response time to a pedestrian by 50% as compared to direct vision [44]; mirrors require proper alignment; and increasing the number of mirrors that a driver must scan may increase driver task fatigue and may decrease attention to the forward roadway. New mirror housings can also introduce new visual obstructions.
- Fresnel lenses, or flat ridged lenses, can be placed on or integrated into side windows to allow the driver to perceive VRUs or vehicles immediately near the vehicle sides [14]. This approach is mainly applicable for B-pillar or belt-line visual obstructions, and no studies of their effectiveness in crash mitigation could be identified. Fresnel lenses cannot mitigate the forward blind zones of tall hoods or thick A-pillars.
- 360° or bird's-eye view cameras can be installed by automakers in the console or installed aftermarket to provide the driver a synthetic view of the

surroundings stitched from four cameras on the vehicle (forward, left, right, rear). Only active at low speeds (e.g., less than 6 mph [45]), the display's accuracy can suffer from stitching artifacts at the corners and from a lack of proper camera calibration. Research on the crash mitigation effectiveness of this technology is currently lacking to understand the potential benefit and optimal system configuration.

- Pedestrian automatic emergency braking can augment human driver reactions. However, under a recent NHTSA rulemaking that takes effect in September 2029, these systems will not be required to stop the vehicle from striking a pedestrian at speeds below 10 km/h (6.2 mph) or in turning maneuvers [46].

None of the above countermeasures enable line-of-sight eye contact or gestural communication between a person outside a vehicle and a driver, and may have inherent limitations as compared to direct vision. While high-vision vehicle design remains the primary countermeasure, technology countermeasures such as these may provide additive or complementary crash avoidance benefits, in recognition of which UN Regulation No. 166 adopted in 2022 a technology-agnostic approach for increasing driver awareness of VRUs in proximity of the front and side of passenger cars and light trucks/vans [29].

4.2. Future Work

Additional study is required to evaluate the relationship between vehicle visibility and VRU deaths, and which of these changes are most associated with VRU crashes, controlling for various confounding factors.

We recommend conducting a robust correlation analysis of the size of the blind zone and some normalized measures of injuries and fatalities. To be robust, the analysis should also control for other factors that are changing as the blind zones have increased. Such a correlation analysis could be conducted not only using the occluded ground area reported here, but also by reprocessing images to provide repeatable metrics for blind zones in particular regions of the environment such as the hood, A-pillar, or mirror profile, and at specific heights above the ground plane.

Additionally, we recommend simulator studies followed by field studies of various countermeasures. Such studies could determine the effectiveness of each potential countermeasure at preventing pedestrian strikes in scenarios that have been identified as high risk due to blind zones and other factors.

Finally, this study could be made more robust by using a larger number of standardized measurements of the blind zones of a broader range of years, makes, and models from the last several decades. A more comprehensive study could also conduct multiple vehicle measurements to account for a range of driver heights.

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